Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_full_solver_evolution.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260519_235755_d
- started_at_local: 2026-05-19 23:57:55
- finished_at_local: 2026-05-20 00:30:07
- duration_hhmm: 00:32
- duration_seconds: 1932.106
- seed_offset: 3000
- replicate_label: d
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.086786	0.0	0.055227	0.806609	0.76	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235058	0.054968	0.121317	0.811502	0.4375	False	0.003917	0.004207
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235667	0.054968	0.121542	0.915929	0.445	False	0.010836	0.566615
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235058	0.054968	0.121317	0.803432	0.4425	False	0.003643	0.00337

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235667	0.054968	0.121542	0.901072	0.435	False	0.004143	0.263013
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.235058	0.054968	0.121317	0.840152	0.44	False	0.001958	0.015417

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.086786, family holdout 0.0, combined 0.055227.
- Accepted novelty 0.806609, complexity 0.76, and adaptation efficiency 0.040944.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.811502, complexity 0.4375, and adaptation efficiency 0.007333.
- Last-epoch runtime 79.794325 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.003917, positive scaffolds 0, Pareto runtime inflation 0.004207.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.004207, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.915929, complexity 0.445, and adaptation efficiency 0.000476.
- Last-epoch runtime 65.7419 ms and distance evaluations 3810.75.

- Validation: surviving False, transplant delta 0.010836, positive scaffolds 1, Pareto runtime inflation 0.566615.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.388628, "gap_delta": 0.000224, "runtime_inflation": 0.566615, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.803432, complexity 0.4425, and adaptation efficiency 0.007406.
- Last-epoch runtime 97.795575 ms and distance evaluations 4385.75.
- Validation: surviving False, transplant delta 0.003643, positive scaffolds 0, Pareto runtime inflation 0.00337.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.00337, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.901072, complexity 0.435, and adaptation efficiency 0.059586.
- Last-epoch runtime 164.845125 ms and distance evaluations 7380.75.
- Validation: surviving False, transplant delta 0.004143, positive scaffolds 1, Pareto runtime inflation 0.263013.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.184271, "gap_delta": 0.000224, "runtime_inflation": 0.263013, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.840152, complexity 0.44, and adaptation efficiency 0.014165.
- Last-epoch runtime 158.145275 ms and distance evaluations 7668.75.
- Validation: surviving False, transplant delta 0.001958, positive scaffolds 1, Pareto runtime inflation 0.015417.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.015417, "same_gap_faster": false}

Judge Appendix

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TSP Modular-Operator Discovery Suite Review

Summary of Conditions

- **Best condition for TSPLIB and transfer:** phase7_full_solver_evolution
- **Total conditions:** 7

- **No surviving candidates** under any condition
- **Key metrics consistently show:**
- No held-out TSPLIB gap improvement (gap reductions = 0)
- No family holdout gap improvement (gap deltas = 0)
- Ablation tests indicate disabling or altering operators causes **no degradation**
- Transplant checks mostly show **no positive gap delta**, sometimes even minor regressions
- Pareto tradeoffs show no gap improvement at the same or similar runtimes
- High runtime inflation for candidate pruner operators without meaningful gains
- No positive family-specific signals; no operators improve any family consistently

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Detailed Interpretation

1. Held-Out TSPLIB and Family Holdout Gaps

- All reported operators (perturbation and candidate pruner types) show **zero or no meaningful reduction** in held-out TSPLIB gap (~0.235 baseline).
- Family holdout gaps show no improvement (best family gain = 0), indicating no generalization or transfer benefits.
- Hence, no operator demonstrates effective transfer or family holdout generalization.

2. Transplant and Ablation Checks

- Ablation tests for all modular operators show disabling or simplifying operators **does not increase gap**, implying the operators do not contribute positively.
- Transplant mean gap deltas cluster around zero or small positive values (gap degradation), except some are slightly positive but insignificant.
- The failure to transplant cleanly into alternative scaffolds (notably clustered_local_search) is consistent across operators.
- Runtime increases significantly with candidate pruner operators, but without gap improvement.
- Therefore, no operator survives the transplant or ablation validation criteria for meaningful modularity and reusability.

3. Pareto and Runtime Tradeoffs

- Pareto results show **no better gap at equal or lower runtime** for any modular operator.
- Runtime inflation can be up to ~0.56x (large) for pruning operators without improvement in solution quality.
- No operator shows speedup at equal gap.
- No operators improve the Pareto frontier meaningfully.

4. Operator Core Ideas and Reusability

- Operators target stagnation escape via deterministic perturbation schedules or candidate pruning adapting to instance descriptors.

- Despite reasonable conceptual ideas, consistent failure in validation implies **no reusable operator structure identified**.
- No operator can be claimed as discovered or reusable modular operator according to validation rules.

Conclusion

- **No operators survived** transplant, ablation, held-out gap, or family holdout gap validations.
- The **full solver evolution baseline** condition yields the best TSPLIB and transfer gaps, but is non-modular and cannot be claimed as a reusable operator.
- Modular operators, both perturbations and candidate pruners, show no validated effect on held-out generalization or family transfer.
- No claims about reusable operator structure or algorithm discovery are supported.
- Runtime inflation and transplant failures further weaken the practical value of proposed operators.

Recommendation: Focus future work on stronger validation designs improving candidate operator transfer and ablation signals, especially on held-out TSPLIB and family holdouts.

Key Metrics Summary Table (selected)

Condition	Type	Final TSPLIB Gap	Final Family Gap	Transplant Mean Gap Delta	Ablation Signal (disabled ∆gap)	Surviving Candidate	Runtime Inflation
phase7_baseline_heuristic_only	Baseline	0.235058	0.054968	0.0	N/A	No	0.0
phase7_full_solver_evolution	Full solver	**0.086786**	0.0	0.0	N/A	No	0.0
phase7_modular_operator_evolution	Perturbation	0.235058	0.054968	0.0039	0.0 (no hurt)	No	~0.0042
phase7_modular_operator_random_replay	Candidate Pruner	0.235667	0.054968	0.0108	-0.0002 (disabled slightly better)	No	0.57
phase7_modular_operator_diversity_residual_replay	Perturbation	0.235058	0.054968	0.0036	0.0 (no hurt)	No	0.0034
phase7_modular_operator_compression_pressure	Candidate Pruner	0.235667	0.054968	0.0041	-0.0002 (disabled slightly better)	No	0.26
phase7_modular_operator_pareto_selection	Perturbation	0.235058	0.054968	0.0020	0.0 (no hurt)	No	0.015

Final Verdict

- **No validated modular operator was found.**
- The best transfer and TSPLIB gaps come from a full solver evolution non-modular approach with no positive modular operator transplant.
- Candidate pruning and perturbation modular operators fail ablation and transplant tests and show no transfer improvement.
- **No claims of algorithm discovery or reusable operator structure are supported by the data.**